

L11: Adaptive (or Coevolutionary) Networks

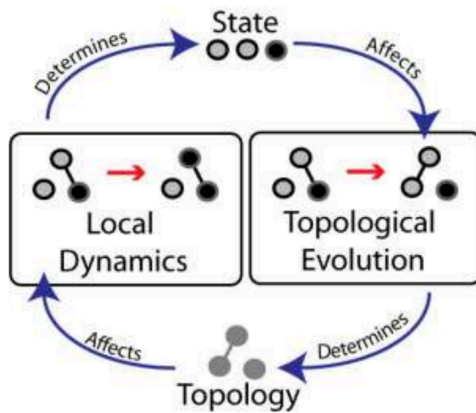


Image Source: "[Adaptive coevolutionary networks: a review](https://arxiv.org/pdf/0709.1858.pdf)"  (<https://arxiv.org/pdf/0709.1858.pdf>) by Gross and Blasius

The most challenging class of problems in network dynamics is related to Adaptive or Coevolutionary networks.

Here, there are two dynamic processes that are coupled in a feedback loop, as shown in this illustration:

- The topology of the network changes over time, as a function of the dynamic state of the nodes,
- The state of the nodes and/or edges changes over time, as a function of the network topology.

In the illustration, we see a small network of three nodes. Each node can be in one of two states: grey and black. The state of a node depends on the state of its neighbor(s). But also whether two nodes will connect or disconnect depends on their current state.

Adaptive network problems are common in nature, society, and technology. In the context of epidemics, for instance, the state of an individual is dynamic (e.g., *Susceptible, Exposed, Infectious, Recovered*), depending on the individual's contacts. That contact network is not static, however – infectious people (*hopefully*) stay in quarantine, while recovered people may return back to their normal social network.

Closing the feedback loop, such changes in the contact network can also affect who else will get exposed.

Another example of an adaptive network is the brain. The connections between our neurons are not fixed. Instead, they change through both short-term mechanisms (such as *Spike-Timing-Dependent Plasticity* or *STDP*) and longer-term mechanisms (such as *axon pruning during development*), and they affect both the strength of existing connections and the presence of synapses between nearby neurons. Importantly, these synaptic changes are largely driven by the activity of neurons. The Hebbian theory of synaptic plasticity, for instance, is often summarized as "**neurons that fire together, wire together**" – a more accurate statement would be that if neuron A often contributes to the activation of neuron B, then the synaptic connection from A to B will be strengthened. The connection from A to B may be weakened in the opposite case.

A key question about adaptive networks is whether the two dynamic processes (*topology dynamics and node dynamics*) operate in similar timescales or not.

In some cases, the topology changes in much slower timescales than the state of the network nodes. Think of a computer network, for instance: routers may become congested in short timescales, depending on the traffic patterns. The physical connectivity of the network however only changes when we add or remove routers and links, and that typically happens in much larger timescales. When this is the case we can apply the "**separation of timescales**" principle, and study the dynamics of the network nodes assuming that the topology remains fixed during the time period that our model focuses on.

In the more challenging cases, however, the two dynamic processes operate in similar timescales and we cannot simplify the problem assuming that either the topology or the state of the nodes is fixed. We will review a couple of such models in the following pages.

A major question in adaptive network problems is whether the network will, after a sufficient time period, "converge" to a particular equilibrium in which the topology and the state of the nodes remain constant. In the language of dynamical systems, these states are referred to as **point attractors** – and there can be more than one. Other types of attractors are also possible, however. In **limit cycles**, for instance, the network may keep moving on a periodic trajectory of topologies and node states, without ever converging to a static equilibrium.

Additionally, fixed points in the dynamics of adaptive networks can be stable or unstable. In the latter, small perturbations in the network topology or state of nodes (*e.g., removing a link or changing the state of a single node*) can move the system from the original fixed point to another.

Food For Thought

Think about the various networks you are familiar with. Are they adaptive or not? Can you study them with the separation of timescales principle? Do you think they have some point attractors or do they exhibit more complex dynamics?